

The effect of tablet error on the precision of estimating the absolute abundance of microfossils in the FOVS method

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2026-06-24

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1 Declaration

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2 Abstract

The FOVS method estimates the absolute abundance of microfossils by using exotic marker tablets. However, the number of markers contained in each tablet varies, which may reduce the precision of the concentration estimates. This study employs repeated simulations to investigate the impact of tablet dose variation on FOVS precision, whether the influence could be mitigated through experimental design, and how the relative importance of tablet errors changes under different conditions. Three scenarios are respectively set up: tablet coefficient of variation (CV), number of tablets, and target-to-marker ratio. Each group of parameters is simulated 5,000 times. The results show that as tablet CV increases, the total theoretical error and the importance of tablet error also increase. Increasing the tablet number can reduce tablet errors and improve estimation precision. The relative importance of tablet errors is also influenced by the target-to-marker ratio: when there are more markers, tablet errors may become the main source of error; when there are fewer markers, the error in marker counting will mask its influence.

3 Introduction

Microfossils can record changes in past biological communities and environments, and thus are often used in the study of ancient environments and ecosystems (Mays, Amores, and Mays (2025),

p. 2). Many studies use relative abundance to describe the proportions of different microfossil types in a sample, but changes in the proportion do not necessarily indicate a change in the actual quantity, because a change in the quantity of one type will also affect the proportion of other types (Jackson (1997), p. 929–932). Absolute abundance can provide the actual number of microfossils in a single sample, thereby enabling a more direct comparison between different samples (Mays, Amores, and Mays (2025), p. 2). However, ordinary quadrat sampling can only observe a small portion of the area on the prepared slides, making it difficult to directly convert the counts in the field of view into the absolute abundance in the original sample (Mays, Amores, and Mays (2025), p. 7). This can be achieved by adding a known number of exotic markers before sample processing, and then determining the conversion by the ratio of the target microfossils to the markers (J. Stockmarr (1971), p. 619). The FOVS method, based on this approach, divides the counting of targets and markers into two stages to enhance the efficiency of data collection (Mays, Amores, and Mays (2025), p. 6–8).

In practical applications, markers are usually introduced by tablets, but the number of markers in each tablet is not exactly the same and there are variations (J. Stockmarr (1973), p. 89; Maher (1981), p. 158–159). This tablet dose variation causes the actual marker dose to deviate from the nominal dose used in the calculation, which may affect the precision of the absolute abundance estimation. Therefore, this study focuses on tablet dose variation, and mainly examines the following three questions: How does tablet dose variation affect the precision of absolute abundance estimation in FOVS method? If the effect is substantial, can it be reduced through changes in experimental design? How does this impact of tablet dose variation vary under different experimental conditions? To address these questions, this study uses simulated data to establish three scenarios that change the marker dose variation in the tablets, the number of tablets added, and the target-to-marker ratio in the sample. For each scenario, we compare the final concentration estimate and its theoretical error. The total error is further separated into its different components for analysis, and the effort required to complete the counts is also recorded.

4 Background

4.1 Exotic marker method

The concentration of microfossils is usually expressed as the number of target particles per unit mass, area or volume of the sediment (Mays, Amores, and Mays (2025), p. 3, Eqn. 1). Since the sediment needs to undergo experimental treatment, and a portion of the treated material is then used to make microscope slides, the particle distribution and concentration on the slides are no longer the same as those in the original sediment. Therefore, ordinary quadrat sampling can only estimate the particle density on the slide, but it cannot separately convert this density into the concentration in the original sample (Mays, Amores, and Mays (2025), p. 7).

However, using exotic markers can solve this problem. The method of calculating pollen and spore density by adding a known amount of exogenous pollen or spores to the sample can be traced back to Benninghoff (1962). First, measure the mass, area or volume of the original sample, and then add a known number of exotic markers. The processed samples are made into slides, and then the target particles and markers are counted simultaneously under the microscope. Since the number of markers added to the sample is known, and this method assumes that the targets and markers

are fully mixed in the treated sample and are similarly affected during the treatment process, the target-to-marker ratio observed on the slide can approximately represent the proportion in the entire sample (Mays, Amores, and Mays (2025), p. 2–3). This is thereby used to estimate the concentration of the target particles in the original sample.

Following Mays, Amores, and Mays (2025) (p. 3, Eqn. 1), the concentration of target microfossils is calculated as:

$$c = \frac{xN_1\bar{Y}_1}{nV}, \quad (1)$$

where, c represents the concentration of the target in the original sample; x is the number of targets counted; n is the number of markers counted; N_1 is the number of doses (e.g., tablets) of markers added to the sample; $\bar{Y}_1$ is the average number of markers per dose; V is the total size (mass, area or volume) of sample. Therefore, $\frac{x}{n}$ represents the target-to-marker ratio within the counting area, while $N_1\bar{Y}_1$ is the nominal total number of markers added to the sample. When these two values are combined, the relative counts on the slide can be converted into the concentration of the original sample.

4.2 Tablet and tablet error

J. Stockmarr (1971) introduced tablets containing a known quantity of spores from the genus *Lycopodium* into absolute pollen analysis, enabling exotic marker particles to be added to the samples at a standardized dose. Each tablet is composed of a certain number of *Lycopodium* spores and a soluble carrier; after the addition of the sample, the carrier dissolves, while the spores, as identifiable exotic markers, participate in the subsequent experimental processing and microscopic counting. Since the spores of *Lycopodium* are usually easily distinguishable from the target microfossils in the sample, and can undergo the same chemical treatment and preparation process as the target particles, the tablets not only provide an internal standard for concentration calculation, but also can to some extent reflect the loss of particles during the experimental treatment process (J. Stockmarr (1971), p. 615–619; Mays, Amores, and Mays (2025), p. 2–3).

However, the actual number of spores in each tablet is not exactly the same. A tablet batch can therefore be described using its mean marker dose and standard deviation. Therefore, in the concentration formula, $N_1\bar{Y}_1$ represents the nominal value, and the actual total number of markers added may be higher or lower than this value (J. Stockmarr (1973), p. 89, Fig. 2; Maher (1981), pp. 158–159).

The random variation between this actual marker dose and its nominal value is referred to as tablet error in this study. Tablet error is measured as the coefficient of variation of marker dose per tablet:

$$CV_T = \frac{\sigma_T}{\bar{Y}_1}, \quad (2)$$

where, σ_T represents the standard deviation of the number of markers per tablet, and $\bar{Y}_1$ is as in Eqn (1).

4.3 FOVS method

In the normal exotic marker method, the slide is typically scanned continuously while both targets and markers are counted until a predetermined number of particles of one type is reached. When the number of targets and markers is relatively close, this method is more effective; however, when the number of targets is much greater than that of markers, it is necessary to count a large number of targets in order to observe a sufficient number of rare markers (Price et al. (2016), p. 78–79, 86 and 89, Fig. 3). At this point, further increasing the target count offers limited assistance in improving precision, but it will increase the workload required for microscope counting.

Mays, Amores, and Mays (2025) proposed the field-of-view subsampling (FOVS) method, which divides the counting of targets and markers into two stages. First, perform the calibration count (Mays, Amores, and Mays (2025), p. 7): Count all the targets in selected microscope fields of view and estimate the average number of targets for each field of view. For example, if there are 20 calibration fields and 100 targets are counted within them, the average number of targets per field is 5. Then, perform the extrapolation count (Mays, Amores, and Mays (2025), p. 7–8): Observe a large number of new fields, and mainly count the markers. In this way, targets do not need to be counted individually in each extrapolation field. Instead, the average value obtained from the calibration count can be used for estimation.

The total number of targets in the extrapolation fields is estimated by (Mays, Amores, and Mays (2025), p. 7, Eqn. 3)

$$\hat{x} = \bar{Y}_x N_E, \quad (3)$$

where, $\hat{x}$ is the estimated total number of targets in the extrapolation fields; $\bar{Y}_x$ is the average number of targets per calibration field; N_E is the number of extrapolation fields. Replacing the direct target count x in the Equation (1) with $\hat{x}$, we can obtain the FOVS concentration estimate (Mays, Amores, and Mays (2025), p. 8, Eqn. 4):

$$c_{F_x} = \frac{\hat{x} N_1 \bar{Y}_1}{nV}, \quad (4)$$

where, c_{F_x} represents the concentration of target microfossils obtained by the FOVS method; $\hat{x}$ is as in Eqn (3); n is the number of markers counted in the extrapolation count; N_1 , $\bar{Y}_1$, V are as in Eqn (1).

4.4 Error structure in FOVS method

The precision of FOVS concentration estimates is expressed by the total standard error, and a smaller value indicates a higher precision in concentration estimates. The total error of FOVS is composed of three sources: the variation of marker dose in the tablets, the sampling variation of target density in the calibration fields, and the counting variation of marker quantity in the extrapolation count. Assuming that these three sources of error are independent of each other, the total error can be expressed as (Mays, Amores, and Mays (2025), p. 8, Eqn. 5):

$$\sigma_{Fx} = 100 \sqrt{\left(\frac{s_{1P}}{\sqrt{N_1}}\right)^2 + \left(\frac{s_{3P}}{\sqrt{N_C}}\right)^2 + \left(\frac{\sqrt{n}}{n}\right)^2}, \quad (5)$$

where, σ_{Fx} is the total standard error of the mean concentration for the FOVS method (in %); s_{1P} represents the proportional standard deviation of the marker quantity per tablet, corresponding to CV_T in this study; N_1 is as in Eqn (1); s_{3P} is the proportional standard deviation of the target count per calibration field of view; N_C is the number of calibration fields; n is as in Eqn (4).

For clarity of interpretation, the three error components can be expressed separately as:

$$E_T = 100 \frac{s_{1P}}{\sqrt{N_1}}, \quad E_C = 100 \frac{s_{3P}}{\sqrt{N_C}}, \quad E_M = 100 \frac{1}{\sqrt{n}}, \quad (6)$$

where, E_T , E_C and E_M respectively represent the tablet error term, calibration error term and marker counting error term.

4.5 Data-collection effort in FOVS method

However, the improvement in precision should also be balanced against the associated effort. Increasing the number of particles or the number of fields of view usually improves the estimation precision, but it may also increase the time required for microscope observation. Mays, Amores, and Mays (2025) (p. 10–11, Eqn. 7) expressed the FOVS method effort as the sum of the particle-counting effort and the microscope field-of-view moving effort. The formula is

$$e_F = (\omega N_C) + x + (\omega N_E) + n, \quad (7)$$

where, e_F is the FOVS data collection effort; N_C is as in Eqn (5); N_E is as in Eqn (3); x is the total number of targets counted in the calibration count; n is as in Eqn (4); ω is the field-of-view transition effort factor. Following Mays, Amores, and Mays (2025) (p. 11), moving to a new field of view is treated as requiring the same effort as counting two individual particles, so the fov transition effort factor is fixed at $\omega = 2$ in all simulations. Therefore, $(\omega N_C) + x$ represents the effort of the calibration count, and $(\omega N_E) + n$ represents the effort of the extrapolation count.

5 Methods

The current studies have separately examined the marker dose variations of tablet and their statistical properties, and have also established a theoretical framework for FOVS concentration, error and effort. However, the studies reviewed here have not explicitly quantified how tablet dose variation affects precision within the FOVS framework, and it is also unclear at present under what conditions the tablet error will become a significant source of error. Therefore, this study examines tablet CV, tablet number, and target-to-marker ratio in three simulated scenarios to quantify the impact of tablet errors on the total error, and to compare the relative influence of tablet errors with other error sources on the total error.

5.1 Overall simulation framework

This study uses repeated random simulation to investigate the impact of tablet errors on the precision of FOVS concentration estimates.

Each simulation is carried out following the steps below:

- Step 1: The simulation area is a square with a side length of 1. Each simulation generates 10,000 targets. The x-coordinate and y-coordinate of each target are independently randomly generated between 0 and 1. Therefore, each position within the area has an equal chance of being selected.
- Step 2: For each tablet, the actual marker dose is independently generated from a normal distribution with a mean of $\bar{Y}_1$ and a standard deviation of $\bar{Y}_1 CV_T$, based on the relationship defined in Equation (2). The generated dose is rounded to the nearest integer and is limited to at least 1 marker. Then, the actual doses of all the tablets are added together to obtain the total number of markers actually added in this simulation.
- Step 3: Based on the actual total number of markers obtained from this simulation, randomly generate the positions of the markers within the same square area. The marker and the target are independent of each other, and each position within the area has an equal chance of being selected.

(See “Simulation settings and target generation” and “Tablet dose variation and marker generation” in Appendix A.1 and A.2. These steps were implemented in the function `simulate_fovs_once()` on lines 6–44 of `sims_Fossil.R`. The simulation parameters and target locations are defined on lines 6–28, while the nominal marker dose, random tablet doses, actual marker total, and marker locations are generated on lines 30–44.)

- Step 4: Each field of view (FOV) is a square with a side length of 0.1. Therefore, the simulation area is divided into 10×10 FOVs, excluding the outermost ring. Subsequently, a random 7×6 block containing 42 candidate FOVs is then randomly selected from this internal grid. From these candidates, 20 FOVs are randomly selected without replacement for the calibration count and another 20 non-overlapping FOVs for extrapolation counts.
- Step 5: Count the targets in each of the 20 calibration FOVs. Particles that fall on the left or bottom boundary of the FOV are counted, while those that fall on the right or top boundary are not counted to avoid double-counting of particles located on adjacent boundaries. Then estimate the average value of the targets in each FOV.
- Step 6: Count the markers in 20 extrapolation FOVs. As with the calibration count method, the left and lower boundaries are included, while the right and upper boundaries are excluded.

(See “FOV grid construction and sampling design” and “Target and marker counting” in Appendix A.3 and A.4. The FOV grid and the non-overlapping calibration and extrapolation fields were generated on lines 46–104 of `sims_Fossil.R`. Target and marker particles within the selected FOVs were counted on lines 106–128 of the same file.)

- Step 7: Using the Equation (3), the total number of targets within 20 extrapolation FOVs is estimated. Calculate the FOVS concentration using the Equation (4), and the nominal marker dose, $N_1 \bar{Y}_1$, is used rather than the simulated actual marker total.
- Step 8: Each set of parameters is simulated 5,000 times, and then the average values of each output are calculated.

(See “Concentration, effort and error calculation” in Appendix A.5 and “Repeated simulations” in Appendix B.1. The concentration estimate, data-collection effort, and theoretical error were calculated in `simulate_fovs_once()` on lines 130–188 of `sims_Fossil.R`. Each parameter setting was repeated 5,000 times using `simulate_fovs_repeated()` on lines 193–220.)

Scenario	Parameter settings
Scenario 1: Tablet CV	<i>Varied:</i> $CV_T = 0, 0.01, 0.03, 0.05, 0.07, 0.09, 0.11$. <i>Fixed:</i> $N_1 = 1, \bar{Y}_1 = 5,000$, and $u = 2$.
Scenario 2: Tablet number	<i>Varied pairs</i> $(N_1, \bar{Y}_1)$: $(1, 5,000), (2, 2,500), (4, 1,250)$, and $(6, 833.33)$. <i>Fixed:</i> $CV_T = 0.04, N_1 \bar{Y}_1 = 5,000$, and $u = 2$.
Scenario 3: Target-to-marker ratio	<i>Varied pairs</i> $(u, \bar{Y}_1)$: $(1, 10,000), (2, 5,000), (4, 2,500), (6, 1,666.67), (8, 1,250), (10, 1,000)$, and $(15, 666.67)$. <i>Fixed:</i> $N_1 = 1$ and $CV_T = 0.04$.
Common settings	$n_{\text{target}} = 10,000, V = 1, N_C = N_E = 20$ and $R = 5,000$.

Table 1: Summary of parameter settings for the three simulation scenarios. The varied and fixed parameters for each scenario are shown.

5.2 Scenario 1: Varying tablet CV

This scenario regards tablet error as an experimental factor, and by altering the tablet CV, we observe its effect on the FOVS precision. In order to systematically compare this effect, seven tablet CV levels are examined, while other parameters remain unchanged. The specific parameter settings are summarized in Table 1. Among them, when tablet CV is 0, it indicates that the marker dose of each tablet is fixed at the average value, without introducing single tablet dose variation. At other levels, the actual marker dose per tablet is randomly generated around the mean value of 5,000 based on the corresponding CV. Each CV level is simulated 5,000 times, while all other parameter settings remain unchanged. Therefore, the differences between the results of each group can be more clearly attributed to the changes in the tablet CV itself.

(See “Tablet CV scenario code and output” in Appendix C.1. The seven tablet CV levels were defined and simulated on lines 278–300 of `sims_Fossil.R`.)

This section is mainly used to answer how the precision of the FOVS concentration estimate will change as the tablet dose variation increases, and whether the influence of the tablet error on the overall error will gradually increase.

5.3 Scenario 2: Varying tablet number

Apart from tablet CV, the second core variable examined in this study is N_1 , which refers to the number of tablets added to the sample. Intuitively, if the dose variations per tablet are independent of each other, then when more tablets are added, the random differences between individual tablets will be partially averaged out at the overall dose level. Therefore, the impact of tablet error on the final concentration estimate should decrease. To verify this, a separate analysis of N_1 is conducted.

In this scenario, tablet CV is fixed at 0.04, which is based on the tablet data reported by J. Stockmarr (1973) (p. 89, Fig. 2). Stockmarr measured 71 tablets from a single production batch and reported the average marker count is 12,488, with a standard deviation of 499 and a corresponding CV of 4.0%. Therefore, in this study it is approximated as 0.04. To isolate the effect of tablet number from the effect of marker abundance, the nominal total marker dose is fixed at 5,000. Therefore, as N_1 increases, the nominal mean marker dose per tablet unit is calculated as:

$$\bar{Y}_1 = \frac{5000}{N_1}. \quad (8)$$

The paired values of N_1 and $\bar{Y}_1$ are summarized in Table 1. This fixed-dose design enables the variation in theoretical error to be attributed to the number of independent tablets, rather than changes in the overall marker abundance.

(See “N1 scenario code and output” in Appendix D.1. The four values of N_1 and their corresponding mean marker doses per tablet were defined and simulated on lines 303–328 of `sims_Fossil.R`.)

This section’s analysis addresses another more practical issue: If tablet errors do indeed affect the FOVS precision, then in the experimental design, could increasing the number of tablets serve as a mitigation method?

5.4 Scenario 3: Varying the target-to-marker ratio

The first two scenarios respectively examine tablet dose variation and tablet number, but the impact of tablet error on the total error might also be influenced by the ratio of targets and markers in the sample. In the FOVS method, estimation precision depends not only on tablet error but also on the number of markers actually counted in the extrapolation counts. If there are fewer markers in the sample, a smaller number of markers will usually be counted during extrapolation counts. A smaller marker count is more susceptible to random fluctuations, thus increasing the marker counting error, and the relative contribution of tablet error in the total error may also change accordingly. Therefore, this study further establishes a target-to-marker ratio scenario to examine the overall error structure under different ratios of targets and markers.

The nominal target-to-marker ratio, u , is defined as:

$$u = \frac{n_{\text{target}}}{N_1 \bar{Y}_1}, \quad (9)$$

where, n_{target} represents the total number of targets, and $N_1 \bar{Y}_1$ represents the nominal total number of markers. Since the actual tablet dose in each simulation can vary randomly, the u here represents the nominal ratio set based on the average marker dose. To obtain these ratios, the mean marker dose per tablet is set as:

$$\bar{Y}_1 = \frac{10000}{u}. \quad (10)$$

Therefore, as shown in Table 1, the paired values of u and $\bar{Y}_1$ are summarized, and the number of targets, tablet number, and tablet CV are held constant, while the nominal marker dose is changed to produce different ratios.

(See “Target-to-marker ratio scenario code and output” in Appendix E.1. The seven ratio levels were defined on lines 331–334 of `sims_Fossil.R`. The corresponding mean marker dose was calculated for each ratio and the repeated simulations were conducted on lines 335–358 of the same file.)

This section mainly examines whether the relative importance of tablet error in total error is constant or whether it changes along with other error terms.

5.5 Error decomposition and output metrics

5.5.1 Error decomposition

This study also conducts an error decomposition analysis for three simulation scenarios. Specifically, for each simulation, the tablet error term, calibration error term and marker counting error term are calculated separately. Then, the average values of each error term from the 5,000 simulations are used for comparison. This approach can further determine which specific source of error is primarily responsible for the change in the total theoretical error, rather than merely assessing whether the overall error has changed.

In order to compare the relative importance of the three error components, their approximate relative contributions to the total theoretical error are also calculated. For example, the approximate relative contribution of the tablet error term is calculated as:

$$S_T = \frac{\bar{E}_T^2}{\bar{\sigma}_{F_x}^2}, \quad (11)$$

where, E_T is as in Eqn (6); $\bar{E}_T$ represents the mean tablet error term; $\bar{\sigma}_{F_x}$ represents the mean theoretical error. The relative contributions of the calibration error term and the marker counting error term are calculated using the same method. The average values of each error component are presented in tables, while the relative contributions are shown in stacked bar charts.

(See “Concentration, effort and error calculation” in Appendix A.5. The tablet, calibration, and marker counting error components and the total theoretical error were calculated for each simulation on lines 153–164 of `sims_Fossil.R`. Their mean values were calculated in `simulate_fovs_repeated()` on lines 224–247 of the same file.)

5.5.2 Output metrics

Let $R = 5000$ represent the total number of simulations, and let r represent an individual simulation. Four mean output metrics are calculated by averaging the corresponding results across all simulations:

$$\bar{\sigma}_{F_x} = \frac{1}{R} \sum_{r=1}^R \sigma_{F_x}^{(r)}, \quad (12)$$

$$\bar{c}_{\text{true}} = \frac{1}{R} \sum_{r=1}^R c_{\text{true}}^{(r)}, \quad (13)$$

$$\bar{c}_{F_x} = \frac{1}{R} \sum_{r=1}^R c_{F_x}^{(r)}, \quad (14)$$

$$\bar{e}_F = \frac{1}{R} \sum_{r=1}^R e_F^{(r)}. \quad (15)$$

- Mean theoretical error, $\bar{\sigma}_{F_x}$: It is the average value of the total theoretical error obtained from 5,000 simulations. For each simulation, $\sigma_{F_x}^{(r)}$ is calculated using Equation (5). It is also the main indicator used in this study to compare the theoretical estimation precision of FOVS.
- Mean true concentration, $\bar{c}_{\text{true}}$: It refers to the average value of the known actual target concentration in the simulation, and is used as a reference point for comparison. In each simulation, the total number of targets is fixed at 10,000, and the sample size, V , is fixed at 1. Therefore, the true concentration, c_{true} , is always 10,000.
- Mean estimated concentration, $\bar{c}_{F_x}$: It is the average value of 5,000 FOVS concentration estimates obtained using Equation (4). It is used to compare with the actual concentration.
- Mean effort, $\bar{e}_F$: It is the average value of the data collection effort obtained by using Equation (7), and is used to compare the relative amount of microscope data collection work required to complete FOVS counting under different parameter settings. This metric determines whether the effect of the simulation conditions on estimation error is accompanied by changes in counting effort.

(See “Repeated simulations” and “Summary output metrics” in Appendix B.1 and B.2. The individual simulation results were combined on lines 193–222 of `sims_Fossil.R`, and the mean concentration, mean theoretical error, mean effort, and other summary statistics were calculated and returned on lines 224–254 of the same file.)

6 Results

6.1 Effect of tablet CV on FOVS theoretical error

The simulation outputs of scenario 1 are summarized in Table 2. This table reports the mean true concentration, mean estimated concentration, mean theoretical total error, and mean effort. The mean total error across the tablet CV levels is presented in Figure 1.

(See “Tablet CV scenario code and output” and “Code for Figure 1” in Appendix C.1 and C.2. The values reported in Table 2 are contained in `summary_cv.csv` and were selected and formatted on lines 29–41 of `analysis_Fossil.R`. Figure 1 was generated from the same results on lines 43–51 of `analysis_Fossil.R`.)

tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error(%)	mean_effort
0.00	10000	10011.09	3.86	3079.19
0.01	10000	10003.99	4.00	3079.84
0.03	10000	10020.99	4.90	3081.06
0.05	10000	10031.62	6.33	3079.91
0.07	10000	10046.27	8.00	3080.23
0.09	10000	10074.64	9.80	3081.11
0.11	10000	10147.97	11.67	3078.35

Table 2: Simulation results across tablet CV levels. The table summarizes the main indicators obtained from 5,000 repeated simulations when the tablet CV increased from 0 to 0.11.

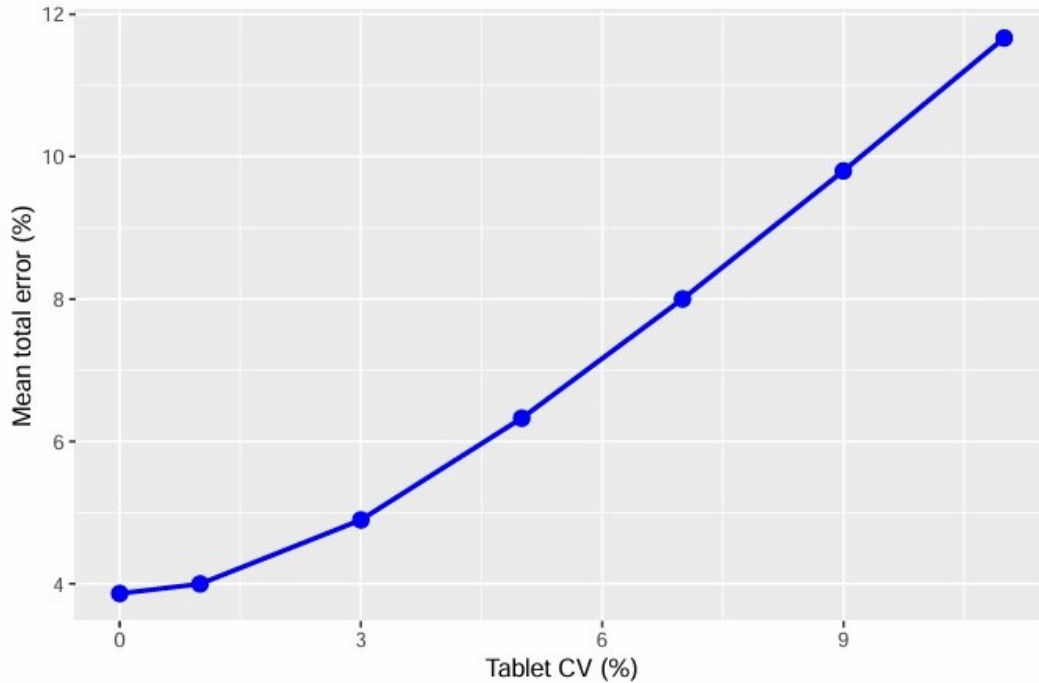


Figure 1: Effect of tablet CV on total error. This graph illustrates how the mean total error of the FOVS concentration estimate changes when the tablet CV increases from 0% to 11%, while all other parameters remain constant.

The corresponding numerical decomposition for scenario 1, including the tablet, calibration, and

marker counting error components, is provided in Table 3. Figure 2 presents the relative contributions of the three error components to the total error.

(See “Error decomposition across tablet CV levels” and “Code for Figure 2” in Appendix C.3 and C.4. The error components reported in Table 3 were obtained from `summary_cv.csv` and formatted on lines 53–62 of `analysis_Fossil.R`. Their relative contributions were calculated on lines 65–70, reshaped on lines 82–93, and plotted in Figure 2 on lines 95–102 of the same file.)

tablet_cv	mean_tablet_term(%)	mean_calibration_term(%)	mean_marker_term(%)	mean_theoretical_error(%)
0.00	0	2.20	3.16	3.86
0.01	1	2.21	3.16	4.00
0.03	3	2.21	3.16	4.90
0.05	5	2.21	3.17	6.33
0.07	7	2.21	3.17	8.00
0.09	9	2.20	3.17	9.80
0.11	11	2.20	3.18	11.67

Table 3: Error decomposition across tablet CV levels. This table reports the tablet, calibration, and marker counting error components and the resulting mean theoretical total error for each CV.

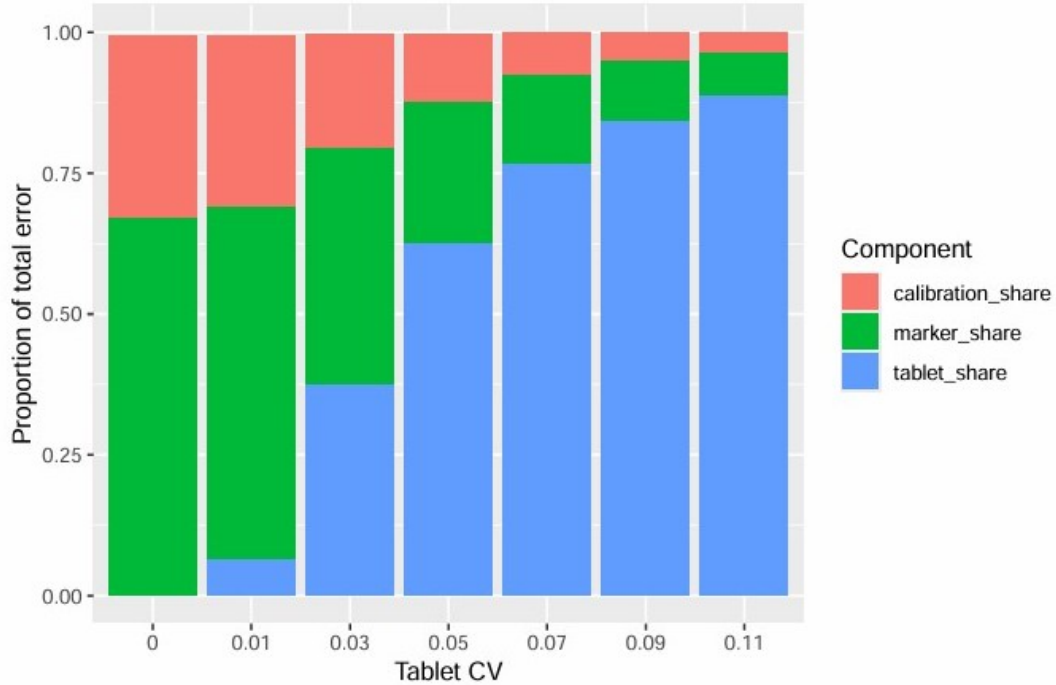


Figure 2: Relative contributions of error components across tablet CV levels. This graph illustrates the proportion of tablet, calibration and marker counting error in the total error as the tablet CV changes.

6.2 Effect of tablet number on tablet and total error

The simulation outputs of scenario 2 are contained in Table 4. The table reports the mean true concentration, mean estimated concentration, mean theoretical total error, and mean effort. The trend of the mean total error varying with N_1 is plotted in Figure 3.

(See “ N_1 scenario code and output” and “Code for Figure 3” in Appendix D.1 and D.2. The values reported in Table 4 are contained in `summary_N1.csv` and were selected and formatted on lines

107–121 of `analysis_Fossil.R`. Figure 3 was generated from the same results on lines 123–131 of the same file.)

N1	tablet_cv	tablet_mean	mean_true_conc	mean_estimated_conc	mean_theoretical_error(%)	mean_effort
1	0.04	5000.00	10000	10032.98	5.57	3078.21
2	0.04	2500.00	10000	10016.91	4.80	3079.84
4	0.04	1250.00	10000	10014.14	4.36	3079.77
6	0.04	833.33	10000	10007.29	4.20	3079.42

Table 4: Simulation results across tablet number. This table summarizes the main indicators obtained from 5,000 repeated simulations for $N_1 = 1, 2, 4, 6$.

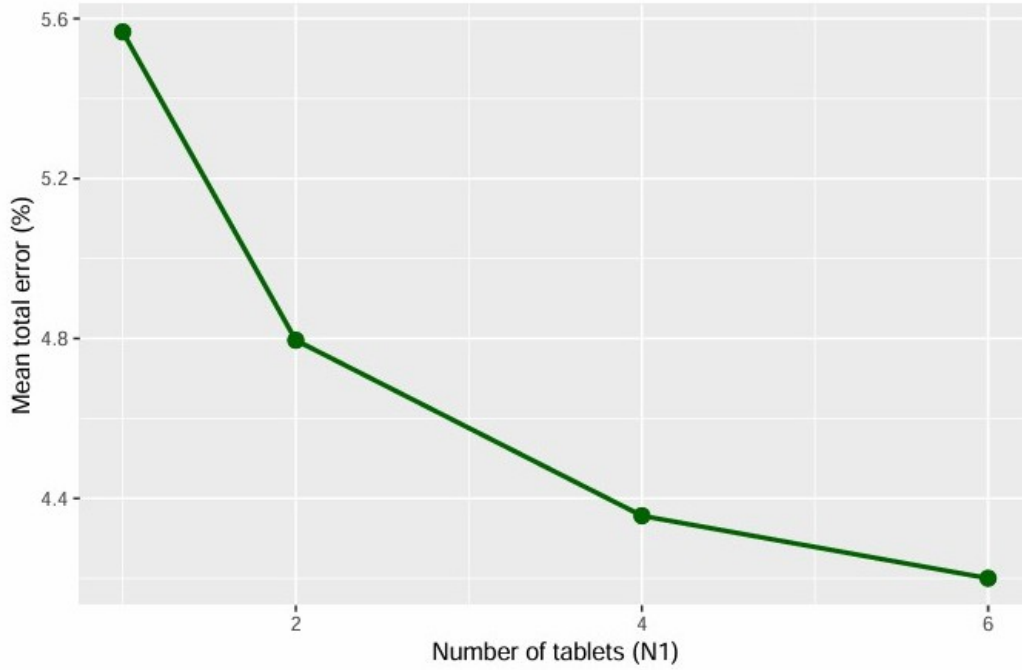


Figure 3: Effect of number of tablets on total error. This graph illustrates how the mean total error of the FOVS concentration estimate changes when N_1 increases, while all other parameters remain constant.

The numerical decomposition of the total error for scenario 2, including the tablet, calibration, and marker counting components, is contained in Table 5. The relative contributions of these components are presented in Figure 4.

(See “Error decomposition across numbers of tablets” and “Code for Figure 4” in Appendix D.3 and D.4. The error components reported in Table 5 were obtained from `summary_N1.csv` and formatted on lines 133–142 of `analysis_Fossil.R`. Their relative contributions were calculated on lines 144–149, reshaped on lines 161–172, and plotted in Figure 4 on lines 174–181 of the same file.)

6.3 Effect of target-to-marker ratio on the error structure

The simulation outputs of scenario 3 are contained in Table 6. The corresponding mean total errors across target-to-marker ratio are plotted in Figure 5.

N_1	mean_tablet_term(%)	mean_calibration_term(%)	mean_marker_term(%)	mean_theoretical_error(%)
1	4.00	2.20	3.17	5.57
2	2.83	2.21	3.16	4.80
4	2.00	2.21	3.16	4.36
6	1.63	2.21	3.16	4.20

Table 5: Error decomposition across numbers of tablets. This table reports the tablet, calibration, and marker error components and the resulting mean theoretical total error for each N_1 .

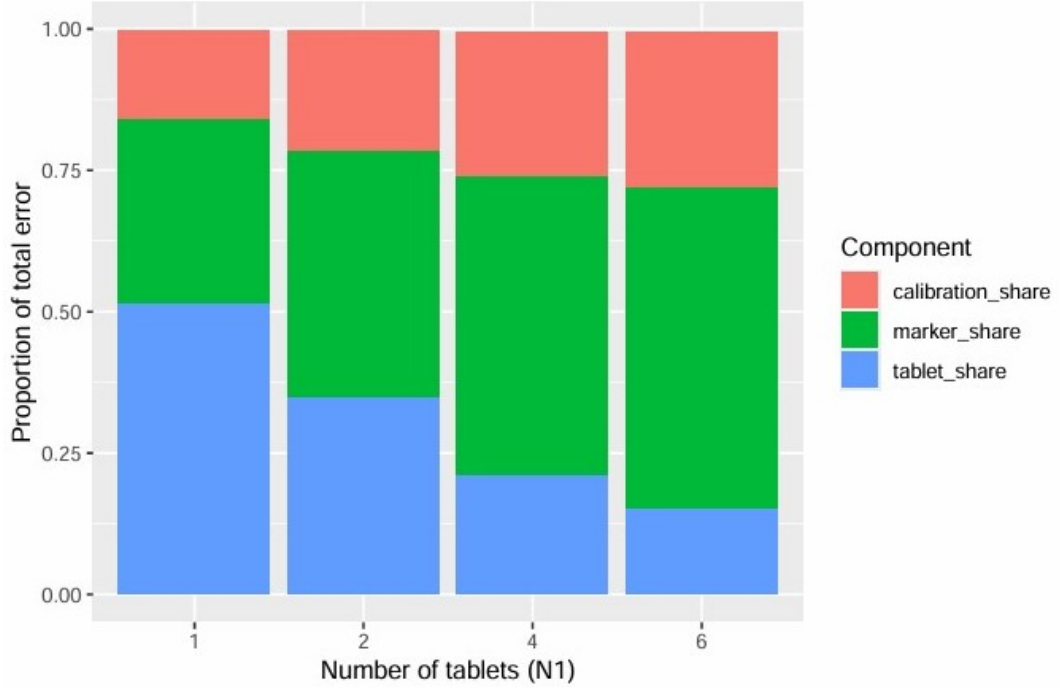


Figure 4: Relative contributions of error components across numbers of tablets. This figure shows the proportion of tablet, calibration and marker counting error in the total error as N_1 varies.

(See “Target-to-marker ratio scenario code and output” and “Code for Figure 5” in Appendix E.1 and E.2. The values reported in Table 6 are contained in `summary_ratio.csv` and were selected and formatted on lines 187–202 of `analysis_Fossil.R`. Figure 5 was generated from the same results on lines 204–212 of `analysis_Fossil.R`.)

ratio	N_1	tablet_cv	tablet_mean	mean_true_conc	mean_estimated_conc	mean_theoretical_error(%)	mean_effort
1	1	0.04	10000.00	10000	10026.91	5.10	4078.43
2	1	0.04	5000.00	10000	10018.62	5.57	3079.36
4	1	0.04	2500.00	10000	10025.48	6.41	2579.89
6	1	0.04	1666.67	10000	10038.58	7.15	2412.98
8	1	0.04	1250.00	10000	10043.79	7.82	2330.57
10	1	0.04	1000.00	10000	10068.92	8.44	2279.91
15	1	0.04	666.67	10000	10090.61	9.83	2212.30

Table 6: Simulation results across target-to-marker ratio. This table summarizes the main indicators obtained from 5,000 repeated simulations for target-to-marker ratios of 1, 2, 4, 6, 8, 10, and 15.

The numerical decomposition of the total error for scenario 3, including tablet, calibration, and marker counting components, is contained in Table 7. The relative contributions of these components are presented in Figure 6.

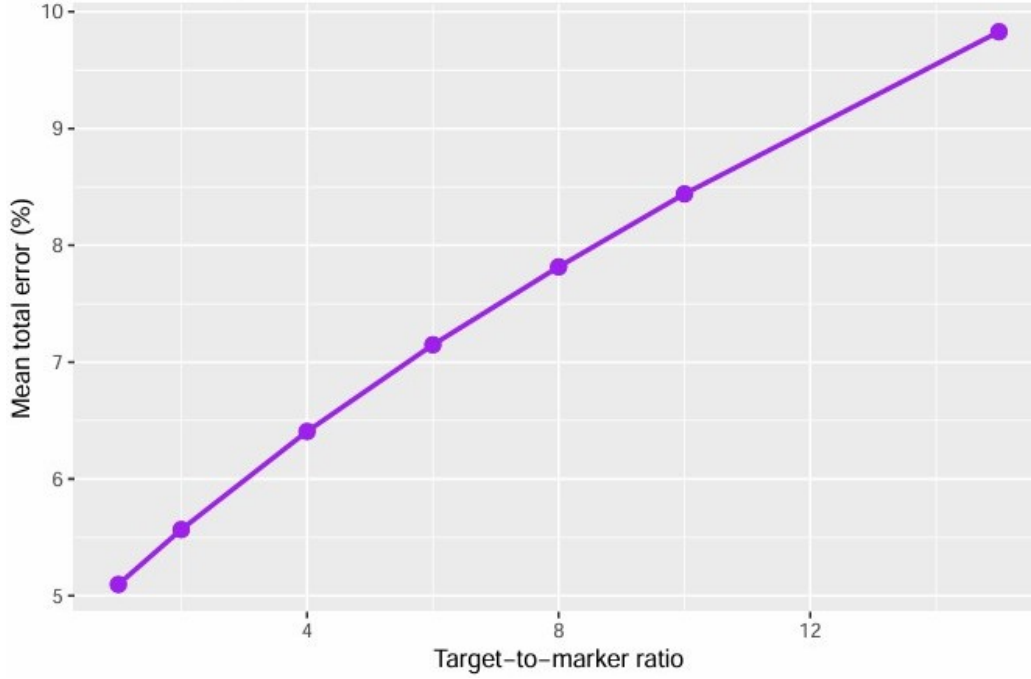


Figure 5: Effect of target-to-marker ratio on total error. This graph illustrates how the mean total error of the FOVS concentration estimate changes when the target-to-marker ratio varies, while all other parameters remain constant.

(See “Error decomposition across ratio” and “Code for Figure 6” in Appendix E.3 and E.4. The error components reported in Table 7 were obtained from `summary_ratio.csv` and formatted on lines 214–223 of `analysis_Fossil.R`. Their relative contributions were calculated on lines 226–231, reshaped on lines 243–254, and plotted in Figure 6 on lines 256–263 of the same file.)

ratio	mean_tablet_term(%)	mean_calibration_term(%)	mean_marker_term(%)	mean_theoretical_error(%)
1	4.00	2.21	2.24	5.10
2	4.00	2.20	3.16	5.57
4	4.00	2.21	4.48	6.41
6	4.00	2.21	5.49	7.15
8	4.00	2.20	6.33	7.82
10	4.00	2.20	7.09	8.44
15	4.00	2.21	8.69	9.83

Table 7: Error decomposition across target-to-marker ratio. This table reports the mean tablet, calibration, and marker error components and the resulting mean theoretical total error for each target-to-marker ratio level.

7 Discussion

7.1 Tablet CV as a limitation on FOVS precision

Table 2 shows that when the tablet CV increases from 0 to 0.11, the mean theoretical error rises from 3.86% to 11.67%. We can also more intuitively observe this trend from Figure 1, which indicates that the tablet error reduces the precision of the FOVS concentration estimates. Table 3 further indicates that this increase is mainly due to the tablet error term increasing from 0% to

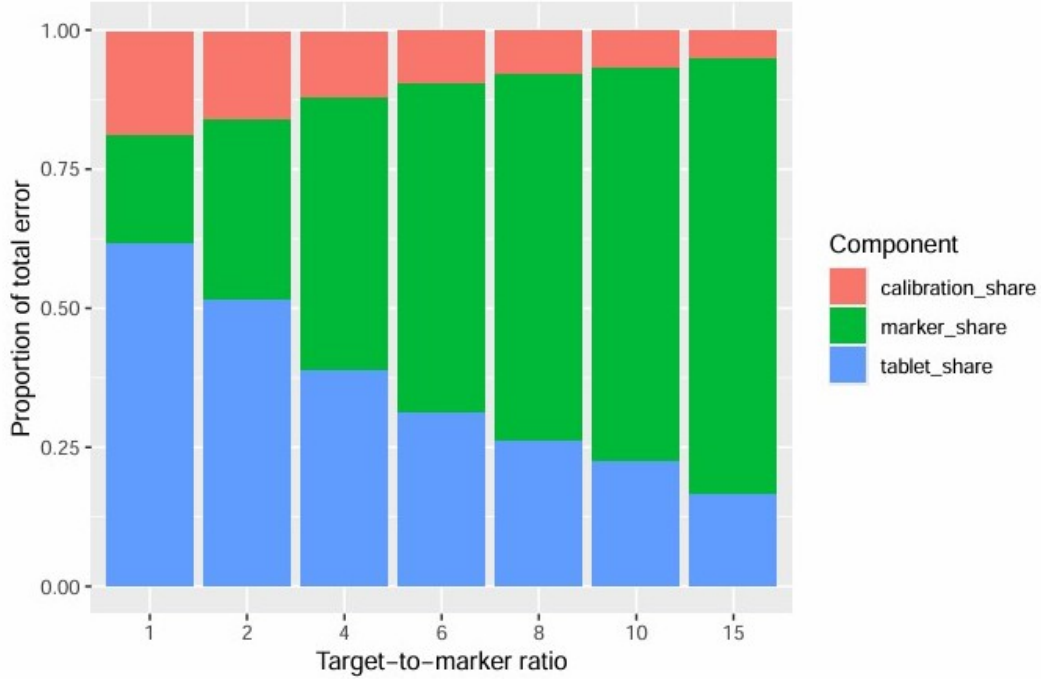


Figure 6: Relative contributions of error components across target-to-marker ratio. This figure shows the proportion of tablet, calibration and marker counting error in the total error as ratio varies.

11.00%, while calibration and marker counting errors remain basically unchanged. Therefore, even if the FOVS counting design is exactly the same, the actual marker dose variation in the tablet may still make the final estimate unreliable.

Figure 2 shows that when the CV reaches 0.05, the tablet error accounts for approximately 62% and has become the main source of error; when the CV is 0.11, this proportion is close to 90%. This means that when the tablet CV is low, the precision can be improved mainly by enhancing the markers and calibration counts. However, when the CV is high, simply increasing the count quantity will have very limited effect, because the main variation has already been introduced by the tablet dose before the counting begins.

Furthermore, the mean estimated concentration shown in Table 2 also slightly increases as the CV increases, which indicates that the tablet error may also cause a certain upward shift in the average result. In conclusion, the increasing CV clearly affects the final estimation precision, and this effect mainly stems from the tablet error term.

7.2 Using multiple tablets to reduce tablet error

Figure 3 shows that when N_1 increases from 1 to 6, the mean theoretical error obviously decreases, from 5.57% to 4.20% (values from Table 4). According to Table 5, the tablet error term decreases from 4.00% to 1.63%, while the calibration and marker counting errors remain largely unchanged. This means that when dose variation among tablets is independent, distributing the marker dose to more independent tablets allows some of the random variation to cancel out, thereby stabilizing the total marker dose.

In other words, if tablet errors have become a main source of errors, then using multiple inde-

pendent tablets might be more reliable than relying on a single tablet. However, Figure 4 shows that as tablet number increases, the contribution of tablet term decreases, while marker counting errors gradually become the largest source of error. Therefore, increasing the tablet number does indeed effectively improve the estimation precision, but it is most helpful only when the tablet error is relatively large. Once marker counting error becomes dominant, further increasing the tablet number provides only limited improvement. At this point, increasing the actual marker count might be more effective.

7.3 Target-to-marker ratio changes the importance of tablet error

Figure 5 shows that as the target-to-marker ratio increases, the mean theoretical error substantially rises, from 5.10% to 9.83% (values from Table 6). The main factor causing this change is the marker counting error. As can be seen from Table 7, the tablet and calibration errors remain relatively stable, but the marker counting error increases from 2.24% to 8.69%.

Figure 6 also shows that when ratio is 1 or 2, tablet error is the main source of errors. This means that when there are already enough markers, adding more markers may not greatly improve precision because tablet error becomes the main limitation. In this case, using tablets with a lower CV or more independent tablets may be more effective. However, as the ratio increases, the relative contribution of tablet error decreases, which does not mean that tablet error itself has been reduced. Instead, the larger marker counting error caused by the fewer markers makes the effect of tablet error less noticeable.

Furthermore, Table 6 also shows that when ratio is 1, mean theoretical error is the lowest; when ratio increases to 2, mean theoretical error only increases from 5.10% to 5.57%, while mean effort decreases from approximately 4,078 to 3,080. Therefore, a ratio of 2 can be regarded as a reasonable balance between precision and effort under the conditions of this study.

7.4 Limitations and future work

This study separately examines the independent effects of tablet CV, tablet number, and target-to-marker ratio. However, in actual experiments, these three factors usually change simultaneously and may also influence each other. Subsequent studies can adopt a comprehensive simulation design and simultaneously compare the different combinations of the three factors, thereby testing whether there are any interactions among them. For instance, the effect of increasing tablet number in reducing tablet error might be more pronounced when the tablet CV is relatively high; however, when the target-to-marker ratio is high and marker counting error becomes the main source of errors, this improvement might be relatively limited. By comparing the total theoretical error, the three error components, and the effort under different combinations, it is possible to further determine under which conditions the tablet error is the most significant. Based on this, an acceptable error level can be set, and the parameter combination requiring the least effort to achieve the corresponding precision could then be identified. This would provide more direct guidance for selecting tablet number and marker dose in experimental design.

8 Acknowledgements

I would like to thank my project supervisor, Anthony Mays, for his guidance and advice throughout this project. I also thank my group members and classmates for helpful discussions, suggestions, and assistance in clarifying some questions during the development of this report. I acknowledge that the structure of this report was prepared with reference to the Part B Report template provided for this course. I acknowledge that I used ChatGPT to help me clarify the easily confused terminology. The code, simulation outputs, and source files supporting this report are available in the following public GitHub repository: https://github.com/FFMMHH-13/MDSresearch_FinalReport_-Fu_-M-.

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10 Statistical appendix

10.1 Appendix A: Core simulation function

10.1.1 Appendix A.1: Simulation settings and target generation

This code defines the basic simulation settings and generates the target particles within the two-dimensional sample region. (Supports Step 1 of Section 5.1, the common simulation settings in Table 1, and Sections 6.1–6.3.)

```
simulate_fovs_once <- function(  
  n_target = 10000,  
  N1 = 1,  
  tablet_mean = 5000,  
  tablet_cv = 0,  
  region_side = 1,  
  fov_size = 0.1,  
  n_cal = 20,  
  n_ext = 20,  
  omega = 2  
) {  
  
  # Sample area  
  V <- region_side^2  
  # True concentration  
  true_conc <- n_target / V  
  # Generate target particles  
  points <- tibble(  
    x = runif(n_target, min = 0, max = region_side),  
    y = runif(n_target, min = 0, max = region_side)  
  )  
}
```

10.1.2 Appendix A.2: Tablet dose variation and marker generation

This code generates the marker particles and introduces tablet dose variation by simulating random marker doses for individual tablets. (Supports Steps 2 and 3 of Section 5.1, the tablet-dose simulation used in Sections 5.2–5.4, and Sections 6.1–6.3.)

```
# Known average marker dose used in the estimator  
known_marker_dose <- N1 * tablet_mean  
# Generate per-tablet marker doses, then sum them  
tablet_sd <- tablet_mean * tablet_cv  
tablet_doses <- rnorm(N1, mean = tablet_mean, sd = tablet_sd)  
tablet_doses <- pmax(round(tablet_doses), 1)  
true_marker_total <- sum(tablet_doses)  
# Generate marker particles
```

```

markers <- tibble(
  x = runif(true_marker_total, min = 0, max = region_side),
  y = runif(true_marker_total, min = 0, max = region_side)
)

```

10.1.3 Appendix A.3: FOV grid construction and sampling design

This code constructs the FOV grid and selects calibration and extrapolation FOVs from a shared local region. (Supports Step 4 of Section 5.1 and the FOV sampling design used in Sections 6.1–6.3.)

```

# Build grid of candidate FOV locations
n_side <- region_side / fov_size
if (abs(n_side - round(n_side)) > 1e-10) {
  stop("fov_size must divide the region side exactly.")
}

all_grid <- expand_grid(
  x = seq(0, region_side - fov_size, by = fov_size),
  y = seq(0, region_side - fov_size, by = fov_size)
)

# Remove outermost edge cells
inner_grid <- subset(
  all_grid,
  x > 0 & x < (region_side - fov_size) &
  y > 0 & y < (region_side - fov_size)
)

if (nrow(inner_grid) < n_cal + n_ext) {
  stop("Not enough inner grid cells for the requested number of FOVs.")
}

# Define a shared local window so calibration and
# extrapolation FOVs come from nearby locations
window_cols <- 7
window_rows <- 6

x_vals <- sort(unique(inner_grid$x))
y_vals <- sort(unique(inner_grid$y))
inner_nx <- length(x_vals)
inner_ny <- length(y_vals)

if (window_cols > inner_nx || window_rows > inner_ny) {
  stop("Shared FOV window is too large for the inner grid.")
}

```

```

start_col <- sample(1:(inner_nx - window_cols + 1), size = 1)
start_row <- sample(1:(inner_ny - window_rows + 1), size = 1)

window_x <- x_vals[start_col:(start_col + window_cols - 1)]
window_y <- y_vals[start_row:(start_row + window_rows - 1)]

shared_grid <- subset(
  inner_grid,
  x %in% window_x & y %in% window_y
)

if (nrow(shared_grid) < n_cal + n_ext) {
  stop("Not enough cells in the shared local FOV region.")
}

# Select calibration and extrapolation FOVs from the same local region
cal_idx <- sample(1:nrow(shared_grid), size = n_cal, replace = FALSE)
cal_fovs <- shared_grid[cal_idx, ]

remaining_grid <- shared_grid[-cal_idx, ]
ext_idx <- sample(1:nrow(remaining_grid), size = n_ext, replace = FALSE)
ext_fovs <- remaining_grid[ext_idx, ]

```

10.1.4 Appendix A.4: Target and marker counting

This code counts target particles in calibration FOVs and marker particles in extrapolation FOVs. (Supports Steps 5 and 6 of Section 5.1 and the target and marker counts used in Sections 6.1–6.3.)

```

# Count fossils in calibration FOVs
cal_fossil <- numeric(n_cal)
for (i in 1:n_cal) {
  cal_fossil[i] <- sum(
    points$x >= cal_fovs$x[i] &
    points$x < cal_fovs$x[i] + fov_size &
    points$y >= cal_fovs$y[i] &
    points$y < cal_fovs$y[i] + fov_size
  )
}

# Count markers in extrapolation FOVs
ext_marker <- numeric(n_ext)
for (i in 1:n_ext) {
  ext_marker[i] <- sum(
    markers$x >= ext_fovs$x[i] &
    markers$x < ext_fovs$x[i] + fov_size &

```

```

    markers$y >= ext_fovs$y[i] &
    markers$y < ext_fovs$y[i] + fov_size
  )
}

```

10.1.5 Appendix A.5: Concentration, effort, and error calculation

This code calculates effort, the concentration estimate, and the three components of theoretical total error. (Supports Step 7 of Section 5.1, Sections 5.5.1 and 5.5.2, and the concentration, effort, and error results reported in Sections 6.1–6.3.)

```

# Calculate effort
x_total <- sum(cal_fossil)
n_count <- sum(ext_marker)
effort <- (omega * n_cal) + x_total + (omega * n_ext) + n_count

# Calculate concentration and error
ybar <- mean(cal_fossil)
xhat <- ybar * n_ext

failed <- (n_count == 0 || ybar == 0)

if (failed) {
  conc <- NA
  tablet_term <- NA
  calibration_term <- NA
  marker_term <- NA
  total_error <- NA
  bias <- NA
  percent_bias <- NA
  s3p <- NA
} else {
  conc <- (xhat * known_marker_dose) / (n_count * V)
  s3p <- sd(cal_fossil) / mean(cal_fossil)

  # Error decomposition
  tablet_term <- 100 * (tablet_cv / sqrt(N1))
  calibration_term <- 100 * (s3p / sqrt(n_cal))
  marker_term <- 100 * (sqrt(n_count) / n_count)

  total_error <- sqrt(
    tablet_term^2 +
    calibration_term^2 +
    marker_term^2
  )
}

```

10.2 Appendix B: Repeated simulation and summary output

10.2.1 Appendix B.1: Repeated simulations

This code repeatedly runs the core simulation and combines the outputs into a single data set. (Supports Step 8 of Section 5.1 and the repeated simulations reported in Sections 6.1–6.3.)

```
simulate_fovs_repeated <- function(  
  n_rep = 5000,  
  n_target = 10000,  
  N1 = 1,  
  tablet_mean = 5000,  
  tablet_cv = 0,  
  region_side = 1,  
  fov_size = 0.1,  
  n_cal = 20,  
  n_ext = 20,  
  omega = 2  
) {  
  results <- vector("list", n_rep)  
  
  for (k in 1:n_rep) {  
    results[[k]] <- simulate_fovs_once(  
      n_target = n_target,  
      N1 = N1,  
      tablet_mean = tablet_mean,  
      tablet_cv = tablet_cv,  
      region_side = region_side,  
      fov_size = fov_size,  
      n_cal = n_cal,  
      n_ext = n_ext,  
      omega = omega  
    )  
  }  
  
  results_df <- bind_rows(results)
```

10.2.2 Appendix B.2: Summary output metrics

This code calculates the summary statistics used in the report, including mean true concentration, mean estimated concentration, mean theoretical error and mean effort. (Supports Sections 5.5.1 and 5.5.2 and the summary statistics reported in Tables 2–7.)

```
mean_conc <- mean(results_df$conc, na.rm = TRUE)  
sd_conc <- sd(results_df$conc, na.rm = TRUE)  
mean_error <- mean(results_df$total_error, na.rm = TRUE)
```

```

sim_cv <- 100 * sd_conc / mean_conc

summary_df <- tibble(
  N1 = N1,
  tablet_cv = tablet_cv,
  mean_true_conc = mean(results_df$true_conc, na.rm = TRUE),
  mean_estimated_conc = mean_conc,
  empirical_sd = sd_conc,
  empirical_cv = sim_cv,
  mean_tablet_term = mean(results_df$tablet_term, na.rm = TRUE),
  mean_calibration_term = mean(results_df$calibration_term, na.rm = TRUE),
  mean_marker_term = mean(results_df$marker_term, na.rm = TRUE),
  mean_theoretical_error = mean_error,
  mean_bias = mean(results_df$bias, na.rm = TRUE),
  mean_percent_bias = mean(results_df$percent_bias, na.rm = TRUE),
  mean_effort = mean(results_df$effort, na.rm = TRUE),
)

```

10.3 Appendix C: Scenario 1 results

10.3.1 Appendix C.1: Tablet CV scenario code and output

This code runs the repeated simulations across different tablet CV levels and produces the corresponding summary results. (Supports the Scenario 1 settings in Section 5.2 and Table 1, and the simulation outputs reported in Section 6.1.)

```

set.seed(456)
tablet_cv_levels <- c(0, 0.01, 0.03, 0.05, 0.07, 0.09, 0.11)

results_cv <- vector("list", length(tablet_cv_levels))

for (i in seq_along(tablet_cv_levels)) {
  results_cv[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = 10000,
    N1 = 1,
    tablet_mean = 5000,
    tablet_cv = tablet_cv_levels[i],
    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

```



```
summary_cv <- bind_rows(results_cv)
```

10.3.2 Appendix C.2: Code for Figure 1

This code generates the figure showing the relationship between tablet CV and mean total error. (Supports Figure 1 in Section 6.1.)

```
ggplot(summary_cv, aes(x = tablet_cv * 100, y = mean_theoretical_error)) +  
  geom_line(linewidth = 1, colour = "blue") +  
  geom_point(size = 3, colour = "blue") +  
  labs(  
    title = "Effect of tablet CV on total error",  
    x = "Tablet CV (%)",  
    y = "Mean total error (%)"  
  )
```

10.3.3 Appendix C.3: Error decomposition across tablet CV levels

This code extracts the three error components for the tablet CV scenario. (Supports Table 3 in Section 6.1.)

```
summary_cv %>%  
  select(  
    tablet_cv,  
    mean_tablet_term,  
    mean_calibration_term,  
    mean_marker_term,  
    mean_theoretical_error  
  ) %>%  
  knitr::kable(digits = 2, caption = "Error decomposition across tablet CV levels")
```

10.3.4 Appendix C.4: Code for Figure 2

This code calculates and plots the relative contribution of each error component across tablet CV levels. (Supports Figure 2 in Section 6.1.)

```
summary_cv <- summary_cv %>%  
  mutate(  
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,  
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,  
    marker_share = mean_marker_term^2 / mean_theoretical_error^2  
  )
```

```

summary_cv_share_long <- summary_cv %>%
  select(
    tablet_cv,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )

ggplot(summary_cv_share_long, aes(x = factor(tablet_cv),
                                   y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across tablet CV levels",
    x = "Tablet CV",
    y = "Proportion of total error",
    fill = "Component"
  )

```

10.4 Appendix D: Scenario 2 results

10.4.1 Appendix D.1: N1 scenario code and output

This code runs the repeated simulations across different values of N_1 . (Supports the Scenario 2 settings in Section 5.3 and Table 1, the simulation outputs reported in Section 6.2.)

```

set.seed(456)
N1_levels <- c(1, 2, 4, 6)

total_marker_dose <- 5000
tablet_mean_levels <- total_marker_dose / N1_levels

results_N1 <- vector("list", length(N1_levels))

for (i in seq_along(N1_levels)) {
  results_N1[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = 10000,
    N1 = N1_levels[i],
    tablet_mean = tablet_mean_levels[i],
    tablet_cv = 0.04,

```

```

    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

summary_N1 <- bind_rows(results_N1)

```

10.4.2 Appendix D.2: Code for Figure 3

This code generates the figure showing the relationship between N_1 and mean total error. (Supports Figure 3 in Section 6.2.)

```

ggplot(summary_N1, aes(x = N1, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "darkgreen") +
  geom_point(size = 3, colour = "darkgreen") +
  labs(
    title = "Effect of number of tablets on total error",
    x = "Number of tablets ( $N_1$ )",
    y = "Mean total error (%)"
  )

```

10.4.3 Appendix D.3: Error decomposition across numbers of tablets

This code extracts the three error components across different values of N_1 . (Supports Table 5 in Section 6.2.)

```

summary_N1 %>%
  select(
    N1,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  ) %>%
  knitr::kable(digits = 2, caption = "Error decomposition across numbers of tablets")

```

10.4.4 Appendix D.4: Code for Figure 4

This code calculates and plots the relative contribution of each error component across the N_1 scenarios. (Supports Figure 4 in Section 6.2.)

```

summary_N1 <- summary_N1 %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )

summary_N1_share_long <- summary_N1 %>%
  select(
    N1,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )

ggplot(summary_N1_share_long, aes(x = factor(N1), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across numbers of tablets",
    x = "Number of tablets (N1)",
    y = "Proportion of total error",
    fill = "Component"
  )

```

10.5 Appendix E: Scenario 3 results

10.5.1 Appendix E.1: Target-to-marker ratio scenario code and output

This code runs the repeated simulations across different values of target-to-marker ratio. (Supports the Scenario 3 settings in Section 5.4 and Table 1, the simulation outputs reported in Section 6.3.)

```

set.seed(456)

ratio_levels <- c(1, 2, 4, 6, 8, 10, 15)
n_target <- 10000

results_ratio <- vector("list", length(ratio_levels))

for (i in seq_along(ratio_levels)) {
  current_ratio <- ratio_levels[i]

```

```

current_tablet_mean <- n_target / current_ratio

results_ratio[[i]] <- simulate_fovs_repeated(
  n_rep = 5000,
  n_target = n_target,
  N1 = 1,
  tablet_mean = current_tablet_mean,
  tablet_cv = 0.04,
  region_side = 1,
  fov_size = 0.1,
  n_cal = 20,
  n_ext = 20,
  omega = 2
)$summary
}

summary_ratio <- bind_rows(results_ratio)
summary_ratio$ratio <- ratio_levels

```

10.5.2 Appendix E.2: Code for Figure 5

This code generates the figure showing the relationship between target-to-marker ratio and mean total error. (Supports Figure 5 in Section 6.3.)

```

ggplot(summary_ratio, aes(x = ratio, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "purple") +
  geom_point(size = 3, colour = "purple") +
  labs(
    title = "Effect of target-to-marker ratio on total error",
    x = "Target-to-marker ratio",
    y = "Mean total error (%)"
  )

```

10.5.3 Appendix E.3: Error decomposition across ratio

This code extracts the three error components across different values of target-to-marker ratio. (Supports Table 7 in Section 6.3.)

```

summary_ratio %>%
  select(
    ratio,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  )

```

```
) %>%
knitr::kable(digits = 2, caption =
  "Error decomposition across
  target-to-marker ratio levels")
```

10.5.4 Appendix E.4: Code for Figure 6

This code calculates and plots the relative contribution of each error component across the target-to-marker ratio scenarios. (Supports Figure 6 in Section 6.3.)

```
summary_ratio <- summary_ratio %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )

summary_ratio_share_long <- summary_ratio %>%
  select(
    ratio,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )

ggplot(summary_ratio_share_long, aes(x = factor(ratio), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components
    across target-to-marker ratio levels",
    x = "Target-to-marker ratio",
    y = "Proportion of total error",
    fill = "Component"
  )
```